Reality Hologram

How does an electron not fall into the nucleus?

Quantum physics postulates that the electron in the stationary state doesn't lose energy, and it rotates around the nucleus forever. Is it quite so? Without receiving some energy of any kind?

The energy is coming from the Sun (more generally, any star) in quanta of energy within a particle, τ neutrino, that provides the continuous movement and even the existence of matter itself. It has greater energy between the neutrinos and generates movement, gravity, and time in the reality that we experience, and that is non-created light. https://michaelvio.github.io/TimpInter.pdf

as in https://michaelvio.github.io/Tint.pdf

Thus, the hydrogen atom's electron must receive energy every $T_0 \approx 6.22 \cdot 10^{-11}$ sec, so the hydrogen atom's electron won't fall on the nucleus. There are 20 quanta of τ neutrino energy that fully charge a Hydrogen's electron with energy in ~1 ns when the hole τ's kinetic energy, about 1.975eV, is transferred to the electron. Hydrogen's electron won't fall on the nucleus. *Every $6{,}22 \cdot 10^{-11}$ sec, the electron charge is changing, forming a dipole and rearranging charge in the atom, thus forming a pair of π^+ and π^- exchanging charge between the electron and the proton.*

But probably the reaction $e + \lceil\tau \xrightarrow{trans} e + \tau\rceil + \pi^+ + \pi^- + \mu\rceil + \Delta E$ (1) is more complex. With a rest mass ν of 0.00038eV/c^2 & Kinetic energy of about 1.6295eV. The tau τ neutrino in the nucleus: we have a τ neutrino that will collide with a pair of a proton and neutron, generating the process of continuous migration of protons, generating gravity, and the μ neutrino that generates the hologram of space https://michaelvio.github.io/ElecFall.pdf:

$p + \lceil\nu \xrightarrow{transf} n + posit\rceil \xrightarrow{decay} (p + e + \nu\rceil) + posit = p + (e + posit) = p$ and equation:

$$p + \lceil\tau \xrightarrow{transf} n + \nu\rceil + \pi^+ + \mu\rceil \xrightarrow{decay} (p + e + \nu\rceil) + \mu\rceil + \lceil\pi^- + \pi^+ = p + (e + \pi^+) = p + \mu + \Delta E_2$$

I guess that they receive quanta of energy for every predetermined period of around ~$6 \cdot 10^{-11}$ sec for a Hydrogen atom at room temperature, 1-S orbital; there is a reaction in which an electron collides with a τ neutrino as in the equation below where monopole is meson, the lightest pion π^+, π^- with a lower rest mass of 0.007eV/c^2. And the τ neutrino will collide with a pair of protons and neutrons, generating the process of continuous migration of protons, generating gravity, and the μ neutrino that generates the hologram of space:

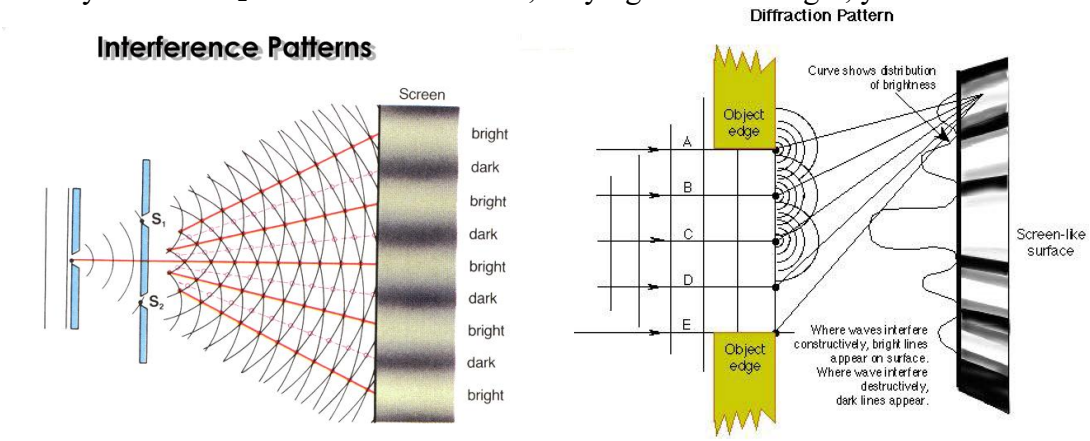
$$p + \lceil\tau \xrightarrow{transf} n + \nu\rceil + \pi^+ + \mu\rceil \xrightarrow{decay} (p + e + \nu\rceil) + \mu\rceil + \lceil\pi^- + \pi^+ = p + (e + \pi^+) + \Delta E \qquad (2)$$

in the Hydrogen nucleus. All the fundamental particles: proton, electron, neutron, photon, magnetron, $Termal_{Particle}$, and phonon interact with τ neutrino and receive a constant amount of energy and temporal ionization. The rest mass of the τ neutrino, produced only in stars, like the Sun, or any star that glows, is around ~0.188 eV/c^2 and that amount is transferred to elementary particles every millionth of a second (microsecond). Thus, we may estimate, with Einstein's equation for τ neutrino of energy ~1.975 eV with rest mass 0.188eV/c^2 so: $E_\tau = \frac{m \cdot c^2}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{0.188eV}{\sqrt{1-\frac{v^2}{c^2}}} = 1.975eV$; V= 0.9954591343·c, thus 99.5459% of the light speed and wavelength ~627nm. In the case of the double-slit experiment, in which an electron beam creates the diffraction pattern on the screen, the application of a beam of light over 627nm wavelength, with the energy support of a τ neutrino, for photons with less than 1.975eV energy, diffraction appears. When observing with photons in the optical window (400-690nm) of the slit that passes the electron doesn't interfere because of blocking the temporal ionization of the electron. Below the τ neutrino kinetic energy Eτ ~ 1.975eV, photons with a wavelength of 627nm, the diffraction will appear. The ratio between the perturbing energy of the photon and the energy of the particle τ neutrino E_τ coming from the Sun is as in the eq: $f + \lceil\tau \xrightarrow{interaction} f + \mu\rceil$. Thus, in the energy balance equation, we approximate the speed as "~c" for all particles: ΔE ≈1.975−1.795 = 0.18 eV for all particles at any Tau neutrino collision and a muon neutrino expulsion. For electrons $e + \lceil\tau \xrightarrow{interaction} e + \mu\rceil$ also ΔE ≈ 0.18eV ± 0.5%.

The threshold values 1.975eV and wavelength ~ 627nm.

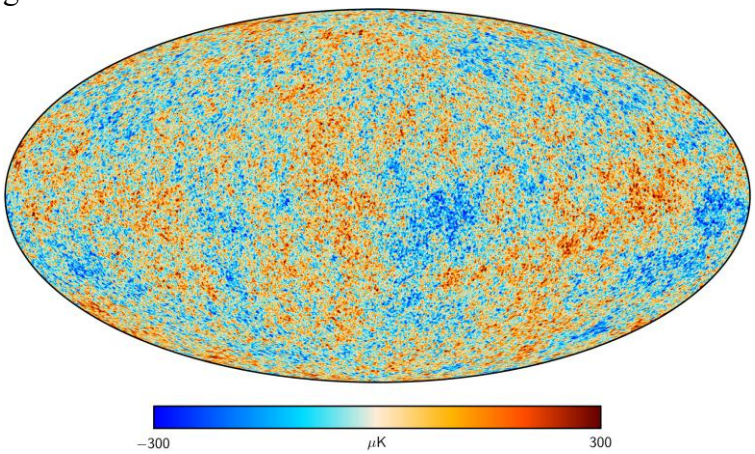
In the double-slit experiment, when an electron beam creates the diffraction pattern on the screen, when a photon beam appears to see which slit the electron passes, the diffraction disappears and a reaction occurs

between the incident "f" photon of energy hν and electron, that blocks the passage of the electron from the wave to the particle during a number N of time quanta T_{qelec}. Thus, observing with a laser beam with a wavelength lower than 627nm will no longer completely block the process of transitioning the particle into a probability wave, so that the interference will no longer appear in the process. This can be verified experimentally with a CO_2 laser of 10.6 microns; varying the wavelength, you can see the exact value.



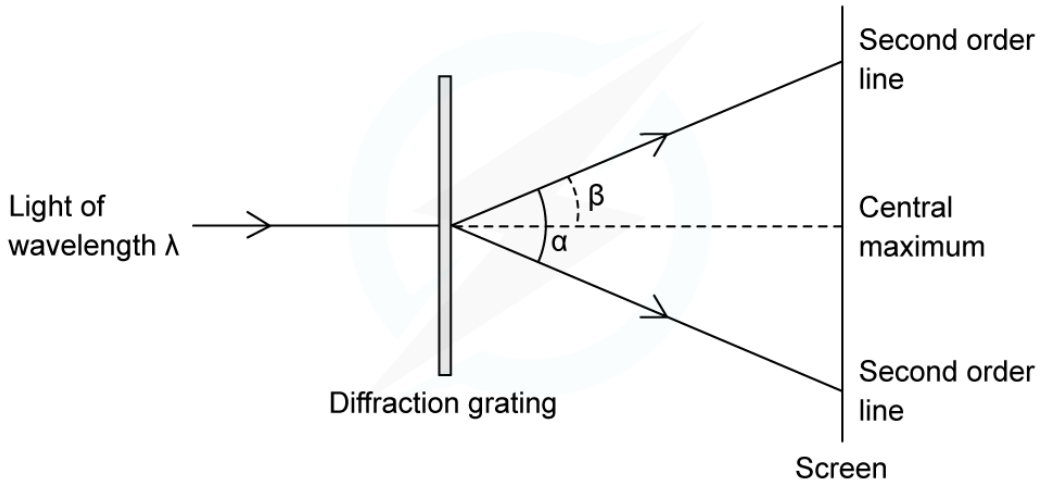
For photons with an energy of approximately 1.975eV (photons with wavelength over 627μm), diffraction appears. The interference does not appear when observing the photon within the optical window, due to the interaction of photons and τ Tau neutrino arriving from the Sun, generating μ muon at the collisions with microparticles.

For photon energies lower than 1.975eV, the diffraction appears, but the threshold value is almost the same wavelength => 627nm for electrons. That's a strange behavior of matter; thus, from another point of view, it's the consequence of the granularity of space-time & mass and the mechanism explained above. The grid of τ neutrinos that are coming from is like the Cosmic Microwave Background because tau neutrinos are coming from all the stars, and we presume that there is a similarity between CMB and the τ neutrino grid.



The temperature fluctuation is less than ~10^{-3}; thus, the variation of τ neutrino flux is similarly probably less than 0.3-0.4%, in any direction.

The kinetic energy of the Tau neutrino (τ) from the Sun is around ~1.975 eV, and its mass ~0.188eV/$c^{2,}$ and we suppose that all the stars that glow emit τ with about the same energy in the range of +/- 0.3-0.4% in any direction. For the experiment, we need a 1-micron step of the diffraction grating and a laser of wavelength 690nm, a pointer of 3-5mW, a control laser beam of 532nm, a pointer of 5mW, and eventually a laser beam of 440nm.



Thus, for the wavelength that produces the resonance between 1 photon and 1 τ neutrino, the optical interference doesn't appear. The interaction of a photon and a τ neutrino involves the wave function of the tau neutrino, and this phenomenon appears at 690nm (photon absorption; thus the energy: $E_\tau = h \cdot c/(10^{-9} \cdot 690)$ (nm) => $E_\tau = 1.975$ eV), so at resonance, the μ neutrinos are less generated. It's a lucky coincidence because the wavelength shouldn't vary by more than one nm, so the phenomenon appears, and on my red pointer write the wavelength 690nm and output 3-5mW. The 1000 lines/millimeters that I bought from Astromagazin https://astromagazin.ro/accesorii-spectroscopie/3113-reea-de-difracie-1000liniimm.html
Thus, the pointer of 532nm is a common one on the market, with a power of 5mW, and you have the picture below. The equation of a transparent grating is $d \cdot sin(\theta_i) - d \cdot sin(\theta_m) = m \cdot \lambda$ (1)
When the planar light wave is normal to the diffraction light has maxima at diffraction angles θ_m given by the diffraction equation as $d \cdot sin(\theta_m) = m \cdot \lambda$ (2)
The grating equation is: $N \cdot \lambda = \sin(\theta)$ where N equals 1000 lines/mm, thus for λ=0.000532mm we have θ = 32.14 degrees, and for 660nm we have θ = 41.22 degrees.
Thus, for m = 2, the second-order is over 90 degrees, and it doesn't exist (sin(θ)>1, absurd).
We estimate from the picture of interference that the angular distance between the three maxima is about ~0.135 radians due to the oscillation of μ neutrinos interacting with photons. We have equation (3)

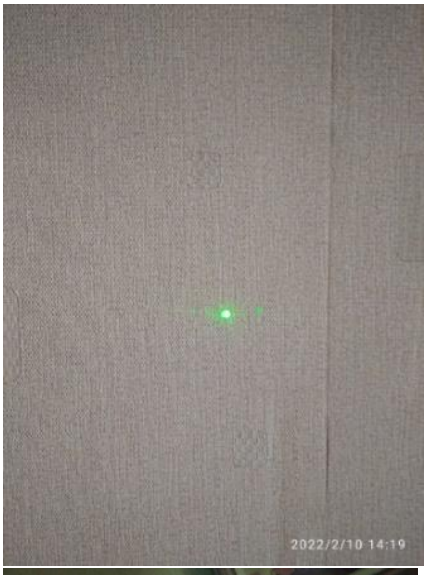
$$\theta_m = arcsin\frac{m \cdot \lambda}{d}$$

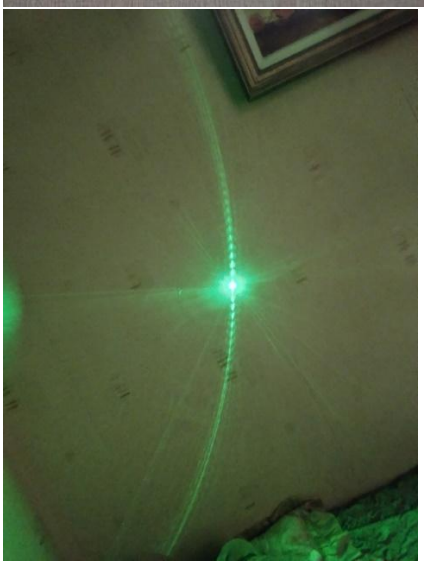
And $\theta_{m1} = 0.135$ degrees, thus ~around 8 minutes of arc (4) $\frac{180\arcsin(0.532)}{\pi} - \frac{180\arcsin(0.534)}{\pi} = -0.135$
Respectively, for $\theta_{m2} = 0.135$ degrees, we have (5) $\frac{180\arcsin(0.536)}{\pi} - \frac{180\arcsin(0.534)}{\pi} = 0.135$
Thus, the decrease in wavelength is less than 2 nm due to the oscillation;
The muon neutrino mass, according to the oscillation observed and measured in other experiments, is around ~0.0088eV/c^2 but the energy associated with the momentum of the muon neutrino is 1.795 eV, and this is the energy of several muon neutrinos that are generated at the incidence between a photon and a τ neutrino.

$$f + \lceil\tau \xrightarrow{interaction} f + \mu\rceil$$

$$e + \lceil\tau \xrightarrow{interaction} e + \mu\rceil \text{ also } \Delta E \approx 0.18eV \pm 0.5\%.$$

This is the qualitative phenomenon; the other approach is in the intensity of the 3 dots on the wall, but I haven't finished yet.
$E_\mu = \frac{m \cdot c^2}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{0.087eV}{\sqrt{1-\frac{v^2}{c^2}}} = 1.795eV$ so $V\mu = 0.9999882542 \cdot c$, thus 99.99% of the light speed and kinetic energy
$E_\mu = 1.795$eV +/-0.3%
https://www.researchgate.net/publication/339913143_DIFFRACTION_GRATING_HANDBOOK_eighth_edition
For the wavelength 532nm, the beam passing through the 1-micron grating produces 3 maximum points, as in the picture:





The wavelength 690nm, the beam passing through the 1-micron grating, didn't produce maximum points as in the picture below:





With a wavelength of 690nm, the beam passing through the 1-micron grating didn't produce maximum points as in the picture above.

The lower images, green and red, are made with the interference produced by a laser & human hair.
https://www.youtube.com/watch?v=v_uBaBuarEM&ab_channel=LookingGlassUniverse

Is reality a Hologram? No, reality is material, but as you can see, almost all microparticles (except probably the particle with quark t = -1 => γ and activates particles) that collide with τ neutrino engulf energy ΔE and emit μ neutrinos, thus related to the Higgs Boson and mass quanta. The multitude of μ particles can create a hologram of reality because there is a one-to-one correspondence between all constituent particles of matter reality and the place where the μ muon neutrino has no space to expand. That place is the closure of the local Universe, or more generally, the closure of the space region if there is a degeneration of space around where there is any kind of degeneration of space.
https://michaelvio.github.io/grating.pdf

The condition is that the incident photon energy should exceed the value of tau τ neutrino energy.

The idea

> Thus, an experiment of double-slit experiment should change its behavior when observed with a wavelength ~532 nm (total energy of the tau neutrino τ => 1.975 eV @ λ = 627 nm) and should be different than observing with 780 ÷ 850 nm. It's simply the energy of interaction with the electron being observed at an energy below 1.6295 eV (total energy of electronic neutrino ν => 1.6295 eV @ λ = 761 nm).

In other words, the model predicts **three distinguishable observation regimes**, set by where the observing photon's energy falls relative to the two neutrino thresholds:

Observing wavelength	Photon energy (hc/λ)	Relative to thresholds	Predicted behavior
~532 nm	≈ 2.33 eV	Above E_τ (1.975 eV) and above E_{ve} (1.6295 eV)	Interaction strong enough to disturb the electron's tau-neutrino recharging — diffraction pattern altered/blocked
~627–761 nm	≈ 1.63–1.98 eV	Between E_{ve} and E_τ	Intermediate regime; the author's earlier tests at 690 nm and 532 nm (see source document) probe this boundary.
~780–850 nm	≈ 1.46–1.59 eV	Below E_{ve} (1.6295 eV)	Too little energy to interact even at the electron-neutrino level — diffraction/interference expected to persist undisturbed

Proposed test

I already used a 1000 lines/mm diffraction grating with 532 nm and 690 nm pointers (3–5 mW) to probe part of this range (see *Reality Hologram*, source document). Extending the same setup to a diode laser in the 780–850 nm band would test the third regime and the specific claim above.

https://www.roithner-laser.com/laser_pointers_780.html?utm_source=chatgpt.com

The mechanism of neutrino generation in matter and solid bodies is the absorption of excitation energy 2·Eν, 2·Eμ, or 2·Eτ in the nucleus and the generation of 2 neutrino particles 2·ν, 2·μ, or 2·τ. So, ionized another nucleus, and an avalanche phenomenon of generation occurs when two particles coherently excite the nucleus, and the nucleus generates 2 identical particles with the same energy. The neutrino, muon, and electron neutrino mechanisms are the same mechanism of action.

Graviton: 1) The 3.259eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.6295eV, i.e., neutrinos with λ= 761nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating antigravity. 2) Theoretically, for photons with λ= 761nm and energy 1.6295eV, after a burst of a few tens to hundreds of nm, it is gravitonically ionized with photons and neutrinos, but not in an avalanche; it can be experimented on sheets (foil) that will levitate. 3) In photons with λ= 253.6 nm and energy 4.8885eV, the generation of 3 neutrinos occurs, which produces levitation at higher light intensity, but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its weight.

Muon neutrino: 1) The 3.59eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.795eV, i.e., neutrinos with λ= 690nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating muon ionization. 2) Theoretically, photons with λ= 690 nm and energy 1.795 eV, after a few tens to hundreds of nm, it is muonically ionized with photons and neutrinos, but not in an avalanche; it can be experimented on sheets (foil) that will enter ionization. 3) With photons with λ= 230 nm and energy 5.385 eV, the generation of 3 neutrinos occurs, which produces ionization at higher light intensity, but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its state.

Tau neutrino: 1) The 3.95eV energy of the photon generates 2 neutrinos of energy 1/2 => 1.975eV, i.e., neutrinos with λ= 627nm, which in turn produce neutrinos with equal energy in the avalanche inside the solid body, generating temporal ionization. 2) Theoretically, for photons with λ= 627nm and energy 1.975 eV, for a few tens to hundreds of nm it is temporally ionized with photons and neutrinos but not in an avalanche; it can be experimented on sheets (foil) that will enter ionization. 3) With photons with λ= 209 nm and energy 5.925eV, the generation of 3 neutrinos occurs, which produces ionization at higher light intensity, but the phenomenon occurs very slowly; the probability of generating 3 neutrinos is lower, with a very precise frequency, but it is detectable; the body changes its state.

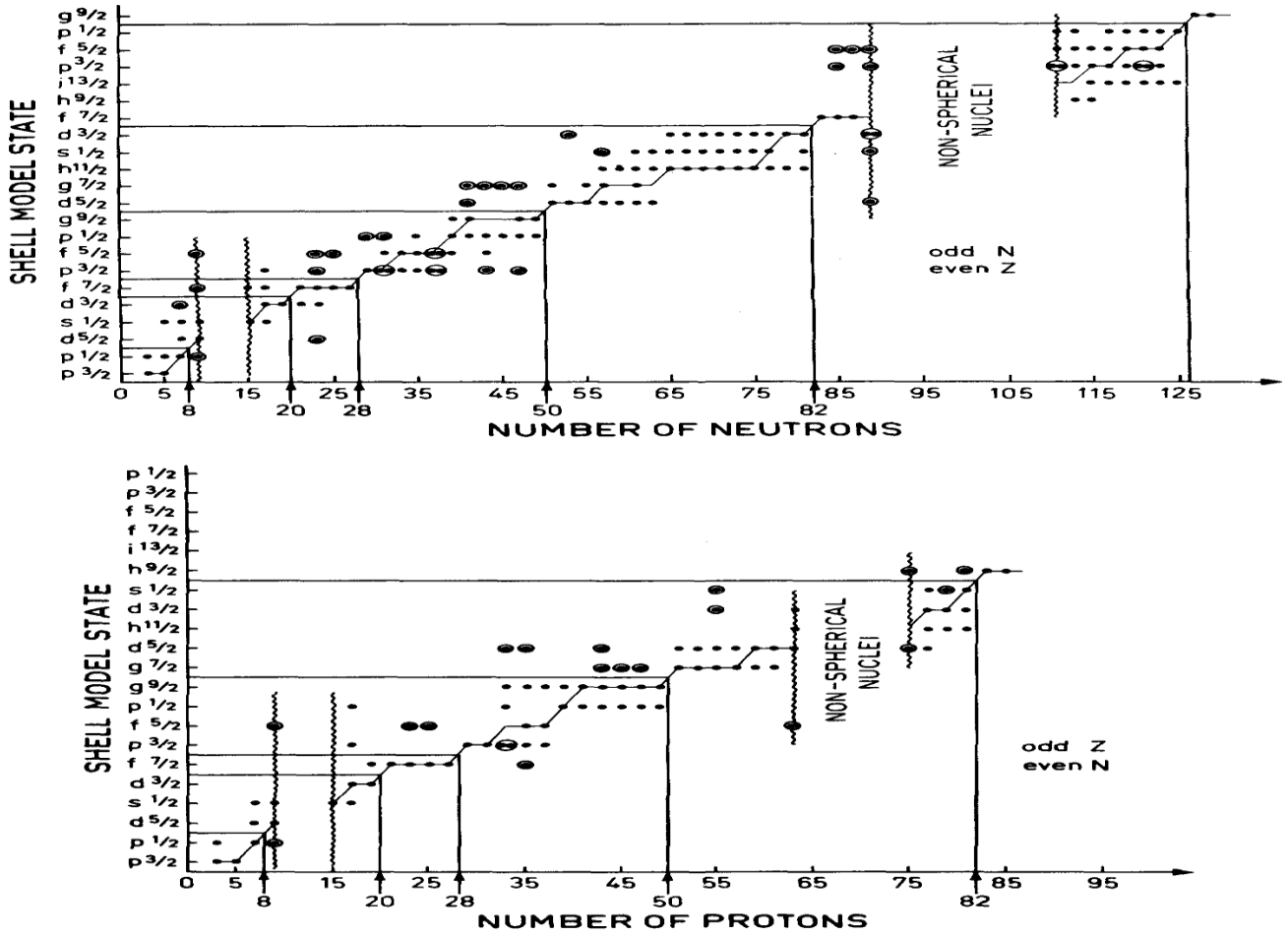


Figure 2-24 *One-particle interpretation of low-energy levels in odd-A nuclei.* The data for the figure are taken from Nuclear Data Sheets. The regions of stable ellipsoidal deformations are excluded, as indicated by the wavy lines. We wish to thank G. T. Ewan for help in the preparation of the figure.

13 Sep. 2026